Jed Renzo Villapando

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EDUCATION

B.Eng in Computer Engineering, *University of Guelph* 09/2024 – present | Guelph, ON
Relative coursework: Engineering and design, Introduction to programming

SKILLS

Languages:
Python, C, Arduino IDE (C++),Java

Tools
AutoCAD, 3D Printing, Excel, Microsoft

PROFESSIONAL EXPERIENCE

Technical Systems Analyst, *Royal Bank of Canada | T&O* 05/2025 – 08/2025 | Mississauga, ON

- **Collaborated** with cross-functional teams to streamline deployment processes, improving release efficiency and **reducing production issues by 20%**.
- **Developed** and **automated** scripts and workflows to support infrastructure deployment, **minimizing manual effort and increasing reliability**.
- **Monitored** and **analyzed** system performance, identifying and resolving recurring issues to **enhance platform stability**.
- **Provided** technical support and troubleshooting for internal tools, **ensuring seamless operations for 500+ users**.
- **Documented** deployment procedures and best practices, **accelerating onboarding and knowledge sharing** within the team.

Programming Lead, *Skills Canada* 09/2022 – 06/2024 | Cambridge, ON

- **Selected from 50 students** to serve as Programming Lead for Skills Canada Competition, representing my high school on a team of 4, and leading us to the **semi-finals at Provincials**.
- **Developed and programmed two advanced robots**, one designed to remove sticks from trees and another to collect and transport balls to a destination.
- **Used Arduino IDE** to program Tetrix circuit components, ensuring efficient power distribution and functionality.
- **Engineered complex drive systems** such as mecanum and omni-directional wheels for precise movement, and programmed Bluetooth connectivity with mapped PlayStation controls for enhanced user experience.

Build & Assembly Lead, *First Robotics Canada* 09/2021 – 06/2024 | Cambridge, ON

- **Directed the construction and mechanical assembly** of an FRC competition robot capable of accurately shooting plastic discs into nets, **leading a team of 6** to achieve **qualification for provincial championships**.
- Strategically assigned roles and enforced **quality control measures**, achieving build precision and maximizing team efficiency.
- **Engineered and wired complex electrical systems** from blueprints, powering key functions such as the shooter, drivetrain, and pneumatic components, enhancing overall robot performance.
- **Leveraged TinkerCAD** for advanced **design and prototyping**, optimizing the development of key components such as the frame, gears, and motors.

PROJECTS

Teddy Bear Wheelchair, *ENGG 1100 Group Term Project* 09/2024 – present

- Designed and programmed a functional robot in a team of 4, using an Arduino board to perform tasks such as launching objects and driving with precision.
- Set up circuits, motor controllers, and sensors, improving **boundary detection accuracy by 100%** and enabling the robot to autonomously navigate and execute its functions effectively.

Tic Tac Toe Game, *Personal Project* 10/2024 – 11/2024

- Developed a digital Tic Tac Toe game as a personal project using the **C programming language**.
- Implemented an **AI feature** that allows the user to play against the computer, enhancing the game's interactivity and complexity.

INTERESTS

E-sports, PC Gaming, Biking, Competitive Basketball